

Name of the Student& USID :	Subject: Ideation And Innovation Challenge (Multi-Disciplinary)
Curriculum : MYP Grade & Section -	Date: 6/6/2026

Dear Students,

Kautuka 3.0 is an innovation-driven platform, designed to explore student's scientific thinking through creativity, investigation, model-making, experimentation, and establishing interdisciplinary connections. The Design Challenge & Ideation Project aims to help students transform real-world problems into workable scientific solutions using observation, imagination, and structured research.

Students From Gr VI to XII may participate **individually or in groups (2–3 members)**.

This year Kautuka 3.0 will be held segment wise

This project aims at:

- Scientific inquiry and innovation
- Design thinking and problem-solving skills
- Collaboration and leadership
- Sustainability awareness
- Art-integrated presentation skills
- Communication and presentation confidence

Project Objectives

Students will be able to:

- Identify a **real-world problem** from their surroundings
- Apply scientific concepts to design a **practical solution or prototype**
- Develop a **model / experiment / simulation / working concept**
- Present findings using charts, data, diagrams, or digital support
- Demonstrate originality, feasibility, and social relevance

Project Themes-

S.No	Project Themes (Students may also propose their own)	Integrated subjects	Grades Involved
01	Agriculture & Food Security	Social /Maths /Lang/Science/AI	(VI to XII)
02	Community Health & Medical Technology	Social /Maths /Lang/Science/AI	(VI to XII)
03	Simple machines in Everyday Life	Science/Social	(VI to XII)
04	Future of Space Science	Social /Maths /Lang/Science/AI	(VI to VIII)
05	Clean Energy Innovations and Environmental Protection Solutions	Social /Maths /Lang/Science	(VIII to XII)
06	Sustainable Living & Smart Resource Use	Social /Maths /Lang/Science/ AI	(VIII to XII)
07	Defence Innovation Challenge: Protect • Detect • Respond Designs related to border security, cyber protection, surveillance, rescue operations, disaster response, or intelligent defence technologies.)	Social /Maths /Lang/Science/AI	(VI to XII)
08	Future Technologies: AI, Robotics & Automation (Concept-based models allowed)	Social /Maths /Lang/Science/AI	(VI to XII)
09	Indigenous Knowledge with relevance to Ancient, Medieval and Modern Science	Social /Maths /Lang/Science/AI	(VI to XII)
10	Assistive Devices for Differently Abled Individuals	Social /Maths /Lang/Science/AI	(VI to XII)

Design Thinking Stages for Students

Students must follow these structured stages while preparing their projects:

Stage 1: Identify the Problem

Ask:

- What issue do you observe around you?
- Who is affected by this problem?
- Why is it important to solve?

Example: Water wastage in school, Plastic usage in households Unsafe, night visibility for cyclists, Crop irrigation challenges etc

Stage 2: Research & Explore

Students should:

- Collect background information
- Study existing solutions
- Identify gaps in current systems
- Record observations in a project log

Sources may include: Books, Teachers, Local observations, Internet, research Field interaction (if possible)

Stage 3: Ideation & Innovation

Students should now:

- Brainstorm multiple solutions
- Select the most feasible idea
- Prepare a rough sketch of their model
- Identify materials required

Encourage ideas that are: Simple, Affordable, Eco-friendly, Locationally relevant, Practical

Stage 4: Model Design / Prototype Development

Students may prepare:

Working model, Static model with explanation, Experimental setup, Digital simulation, Infographic-supported concept model, Art-integrated scientific installation

Models using **recycled or low-cost materials** will receive appreciation.

Stage 5: Testing & Improvement

Students should:

- Check whether their model works
- Modify design if required
- Improve stability and clarity
- Ensure safety and neat presentation

Stage 6: Presentation Planning: Each team must prepare:

- ❑ Project Title
- ❑ Problem Statement
- ❑ Scientific Principle Used
- ❑ Working Explanation
- ❑ Applications
- ❑ Advantages
- ❑ Future Scope
- ❑ Students may present using: Charts /Posters /Data Tables /Graphs /Digital slides short demonstrations

Submission Requirements

Students must submit the following on or before **6th June 26**

1. Project Title Page
2. Problem Statement (1 page)
3. Scientific Concept Explanation (2 pages)
4. Model Sketch / Diagram
5. Materials Used List
6. Working Principle
7. Applications in Real Life
8. Innovation / Improvement Suggested
9. Bibliography / References
10. Team Contribution Sheet (for group projects)
11. Additions: Survey data Photographs of: (i) preparation stages (ii) Prototype testing
(iii) observations (iv) Field interaction notes

Projects will be assessed based on:

Criteria	Description	4M	3M	2M	1M
Originality	Uniqueness of idea				
Scientific Understanding	Concept clarity				
Creativity	Design and presentation quality				
Practical Application	Real-world usefulness				
Sustainability	Eco-friendly approach				
Teamwork	Collaboration effectiveness				
Communication	Explanation clarity				
Model Functionality	Working effectiveness				

Bonus Innovation Indicators (+8 Enrichment Marks) : If a child Integrates with other subjects, SDGs, coding integration, actual feedback evidence provided etc.,

----- Happy Innovation and Ideation -----

